ADEPT — Random Forest Base Learner

Phase 2 Results: Analysis and Next Steps

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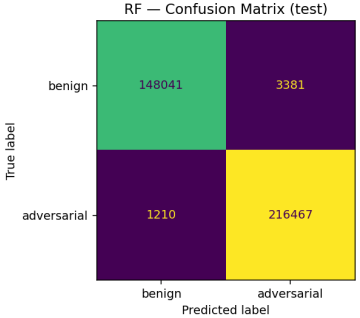
What was trained

- **Task:** binary classification on the 24-dim per-window aggregates from Phase 1 (`rf_stats.npy`). Positive class = TANTGen-perturbed window; negative class = 4 matched benign sources (ddos, dos_slow, mirai, ssdp_flood) — the only attacks for which TANTGen produced structurally valid adversarial PCAPs (portscan / syn_dos excluded: 1–2 packet flows; fuzzing excluded: snaplen no-op).
- **Test split:** flow-atomic, stratified by source (∼369k windows; 59% adversarial / 41% benign — 3 modes per attack make adversarial-majority).
- **Best hyperparameters** (grid search, val macro-F1): `n_estimators=200`, `max_depth=None`, `min_samples_leaf=1`.

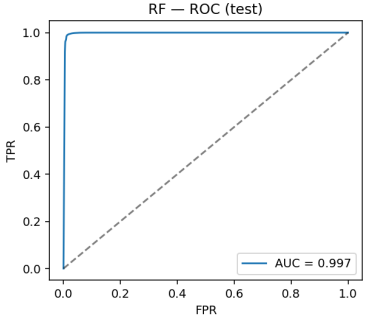
Headline metrics (test, overall)

Macro F1	ROC AUC	PR AUC	Recall (adv)	Precision (adv)
0.987	**0.997**	**0.996**	0.994	0.985

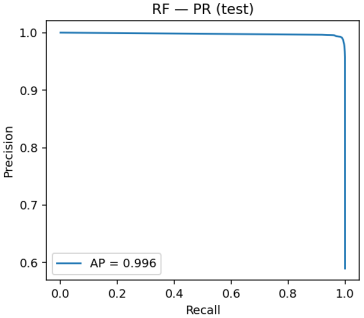
Confusion matrix & curves



Confusion Matrix



ROC (AUC = 0.997)



Precision–Recall (AP = 0.996)

- Strong global separability between benign and TANTGen-adversarial windows.
- PR curve nearly overlapping ROC confirms the 59/41 adversarial-majority imbalance is well handled by the `class_weight=balanced_subsample` setting.

Per-variant performance (vs. balanced benign)

Variant	$F1_{adv}$	ROC AUC	PR AUC
ddos_adv_{both, iat, size}	0.91 / 0.90 / 0.91	0.995–0.998	0.917–0.957
dos_slow_adv_{both, iat, size}	0.955 / 0.953 / 0.955	0.997–0.998	0.977–0.980
ssdp_flood_adv_{both, iat, size}	0.91 / 0.91 / 0.91	0.998	0.959
mirai_adv_both / iat	0.887 / 0.883	0.994–0.995	0.653–0.708
mirai_adv_size	**0.761**	0.983	**0.433**

- **DDoS / DoS Slow / SSDP Flood:** strong detection across all three perturbation modes — these are the variants ADEPT must catch.
- **Mirai:** acceptable on `both`/`iat`; **size-only noticeably weaker** (small payload-padding signal; low PR AUC reveals class-boundary fragility).

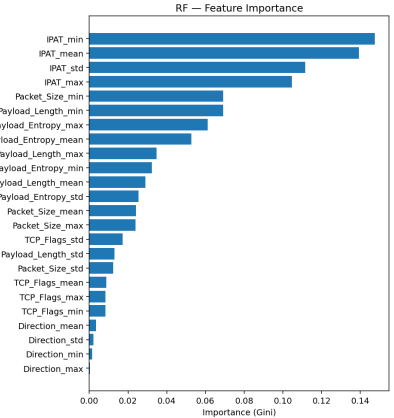
Same-base vs. balanced benign: where detection is harder

Each variant is scored against two benign pools. **Balanced:** K × n_adv windows drawn from all test benign (slide 4). **Same-base:** only benign windows from the *same* source capture.

Variant	F1 (balanced)	F1 (same-base)	Δ
ddos_adv_{both, iat, size}	0.91 / 0.90 / 0.91	1.00 / 0.99 / 1.00	+0.09
dos_slow_adv_{both, iat, size}	0.955 / 0.953 / 0.955	1.00 / 1.00 / 1.00	+0.05
ssdp_flood_adv_{both, iat, size}	0.91 / 0.91 / 0.91	0.99 / 0.99 / 0.99	+0.08
mirai_adv_both / iat	0.887 / 0.883	0.667 / 0.660	**−0.22**
mirai_adv_size	0.761	**0.553**	**−0.21**

- **Volumetric attacks:** same-base F1 *exceeds* balanced F1 — TANTGen's IPAT shifts are so pronounced that detection is easy even against same-source benign traffic.
- **Mirai:** same-base F1 drops by ∼0.21–0.22 — *capture-specific bleed*: same-flow benign and adversarial Mirai windows share underlying timing patterns, making them harder to separate when the benign pool is restricted to the same capture.

Feature importance



Top discriminators

- IPAT statistics (mean / std / min / max) dominate — TANTGen's timing perturbations are the strongest signal.
- Packet_Size and Payload_Length aggregates contribute meaningfully on the `size`/`both` modes.
- Direction and TCP_Flags aggregates — secondary, mostly useful for capture identification rather than perturbation detection.

Implication:

Before training, each of the 24 features is standardized using a mean and standard deviation computed *jointly* from both benign and adversarial windows (`mean_std.json`). This means the RF sees every feature as a z-score — how far a window deviates from the combined population average. Because TANTGen's timing perturbations shift inter-packet arrival times well outside the joint mean, those IPAT z-scores become the dominant discriminative signal for the model.

Conclusions

1. **On the principled 4-attack dataset (ddos, dos_slow, mirai, ssdp_flood), the 24-dim RF reaches 0.987 macro-F1 / 0.996 PR-AUC** — a strong baseline for the statistical half of the ADEPT ensemble.
2. **Volumetric attacks (DDoS, DoS Slow, SSDP Flood) are strongly detected across all three perturbation modes** — same-base F1 $\geq$ 0.99, confirming IPAT shifts are the dominant signal.
3. **The size-only mode is consistently weakest — expected, since RF aggregates compress per-packet padding into mean/std/min/max. Mirai size drops furthest (F1 = 0.76, PR AUC = 0.43).**
4. **Mirai shows capture-specific bleed: same-base F1 drops 0.21–0.22 below balanced F1 — same-flow benign and adversarial windows share timing patterns the RF cannot fully untangle.**
5. **PortScan and SYN_DoS are structurally incompatible with N-packet windowing** (1–2 packet flows yield <200 windows from >150k flows) — excluded from training and evaluation.
6. **Fuzzing adversarials are snaplen-200B no-ops** — excluded from training; their near-zero F1 confirms TANTGen produced no effective perturbation.

Next steps

Short-term (End of May) — complete the ensemble & tune end-to-end

1. **Phase 3 — TCN base learner** on `sequence.npy`. Dilated temporal convolutions should close the gap on `size`-only and Mirai variants where RF aggregates compress the signal.
2. **Phase 4 — Mini-Transformer meta-learner** fusing RF + TCN logits.
3. **Sync with Inês** on the TANTGen version in use — confirm whether the Fuzzing snaplen-200B no-op is intrinsic or a fixable config issue.
4. **DPSS hyperparameter sweep** — script to grid-search the full pipeline: Packet-to-Tensor `window_size` $\in \{5, 10, 20\}$ (smaller N recovers PortScan / SYN_DoS), extended RF grid, TCN & Mini-Transformer params (TBD). Objective: maximize ensemble detection accuracy. Long-running — hence offloaded.

Long-term — defense layer & write-up

5. **Phase 5 — Purification Transformer** (decoder-only Seq2Seq) projecting adversarial windows back onto the benign manifold; KV-caching for latency.
6. **Phase 6 — RL Adaptation Agent** re-weighting the ensemble online from TANTGen feedback + offline DPI validation.
7. **Dissertation write-up.**

Reproducibility

Artifacts under `ADEPT/data/models/rf/`

- `rf.joblib` — fitted classifier (`n_estimators=200`, `max_depth=None`, `min_samples_leaf=1`)
- `metrics.json` — overall + per-source (vs_all / vs_balanced / vs_same_base) + per-`label_str`
- `split.npz`, `dataset.npz` — exact train/val/test partition
- `training_info.json` — full grid-search trace
- `*.png` — confusion matrix, ROC, PR, feature importance

Re-run from scratch

```
python3 scripts/adversarial_generation/generate_adversarials.py
python3 scripts/packet_to_tensor/process_all_attacks.py --tantgen
python3 scripts/rf/run_rf.py
```